Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS, NOVEMBER 2022

DATABASE MANAGEMENT SYSTEMS [ELE 4079]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 21 NOV 2022** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Explain the purpose of database systems. ***(03)***

1B. Illustrate query processing in database systems with the help of a neat sketch. Label all important parts. ***(03)***

1C. What are the conceptual, logical, and physical features identified during database design? ***(04)***

2A. Statement:

"An electricity supply company installs energy meters in the customer premises to record their energy consumption. The customer is billed every month for their energy consumed."

For the statement given above design (draw) an Entity-Relationship (ER) model with proper notations. ***(03)***

2B. With reference to the ER model developed in **Q2A**, for the binary relationship set between electricity supply company and the customer, illustrate the application of different integrity constraints. ***(03)***

2C. With reference to the ER model developed in **Q2A**, assume that there exists a ternary relationship between the electricity company, customer, and energy meter. Without adding any additional entities, show using a suitable ER model, how this ternary relationship can be converted into binary relationship sets. ***(04)***

3A. With the help of suitable examples, explain different anomalies in a relational model. ***(03)***

3B. By applying principles of relational database design and concepts of normal forms, **normalize** the relational model given below. Show the tables created by normalizing "Projects" table. ***(03)***

Projects

ProjectNo	Project Name	Manager	Budget	EmployeeCode	Employee Name	DepartmentNo	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	$100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Ms. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Ms. Charu	500500	$104	Jitendra	D02	Testing	17.00
PC011	HR System	Ms. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance	Mr. Rajesh	710700	$137	Rahul	D03	Database	21.50
PC012	Attendance	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

3C.

Projects

ProjectNo	Project Name	Manager	Budget	EmployeeCode	Employee Name	DepartmentNo	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	$100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Ms. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Ms. Charu	500500	$104	Jitendra	D02	Testing	17.00
PC011	HR System	Ms. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance	Mr. Rajesh	710700	$137	Rahul	D03	Database	21.50
PC012	Attendance	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

With reference to the relational model "Projects" given above, write relational algebraic expressions for the following:

1. Find the names of the employees working in the "Database" department.

2. Find the names of the employees working under manager named "Mr. Ajay".

3. Count the total number of projects.

4. Find the names of the employees working under the same manager. ***(04)***

4A.

Projects

ProjectNo	Project Name	Manager	Budget	EmployeeCode	Employee Name	DepartmentNo	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	$100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Ms. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Ms. Charu	500500	$104	Jitendra	D02	Testing	17.00
PC011	HR System	Ms. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance	Mr. Rajesh	710700	$137	Rahul	D03	Database	21.50
PC012	Attendance	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

With reference to the relational model "Projects" given above, write SQL statements for the following:

1. Create "Projects" table in the database with necessary integrity constraints.
2. Insert the first row of the "Projects" table above into the newly created table in the database.
3. Update the name of the manager "Mr. Ajay" as "Dr. Ajay Sharma". ***(03)***

4B.

Projects

ProjectNo	Project Name	Manager	Budget	EmployeeCode	Employee Name	DepartmentNo	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	$100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Ms. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Ms. Charu	500500	$104	Jitendra	D02	Testing	17.00
PC011	HR System	Ms. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance	Mr. Rajesh	710700	$137	Rahul	D03	Database	21.50
PC012	Attendance	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

With reference to the relational model "Projects" given above, write SQL statements for the following:

1. Find the names of the employees working in the "Database" department.

2. Find the names of the employees working under manager named "Mr. Ajay"

3. Count the total number of projects. ***(03)***

4C. Draw a neat sketch of a state transaction diagram. Explain different transaction states. ***(04)***

5A. List out at least three advantages and disadvantages of importing data in a CSV (Comma Separated Values) format into a MySQL database. ***(03)***

5B. What is XAMPP? Explain the features offered by XAMPP? ***(03)***

5C. Electricity company desires to install smart energy meters at their consumer premises. The company wishes to have a complete design including hardware and software implementation of the smart energy meter.

Suggest the following for the electricity company:

1. A block diagram approach to implement the hardware for measuring energy.
2. List the required software for data acquisition, storage, and display.
3. A sample relational schema showing the data acquired by the sensors.
4. Show the key constraint(s) for the sample relational schema in (3). ***(04)***

Projects

ProjectNo	Project Name	Manager	Budget	EmployeeCode	Employee Name	DepartmentNo	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	$100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
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